Presentation Transcript

Comparative Intelligence and Ethics in Machine Learning Systems

Fashion-MNIST: HOG + Linear SVM vs CNN

Slide 1 — Title

Good day. This presentation is about Comparative Intelligence and Ethics in Machine Learning Systems. I compare a classical computer-vision pipeline with a deep-learning model on the Fashion-MNIST dataset (Xiao, Rasul and Vollgraf, 2017). The aim is not only to ask which model gives the highest accuracy. I also examine how the models generalise, where they fail, what explainability methods reveal, how robust they are to image degradation, and what their errors would mean in a realistic deployment such as an online marketplace.

Slide 2 — Research question and dataset

The research question is: how does a classical machine-learning approach based on engineered visual features compare with a convolutional neural network in terms of classification performance, generalisation, explainability, robustness and real-world suitability? I selected Fashion-MNIST because it is a public benchmark with 70,000 grayscale clothing images, split into 60,000 training images and 10,000 test images. It has ten balanced classes. This balance makes accuracy more meaningful, but the dataset is still challenging because classes such as Shirt, T-shirt/top, Coat and Pullover are visually similar at only 28 by 28 pixels.

Slide 3 — Two representation strategies

The main technical comparison is about representation. In the classical pipeline, the image is first transformed using HOG, or Histogram of Oriented Gradients. HOG describes local edge orientations and shape patterns, and the Linear SVM (Cortes and Vapnik, 1995) then learns decision boundaries in that engineered feature space (Dalal and Triggs, 2005). In contrast, the CNN receives the image directly and learns its own filters during training. This is the difference

between hand-engineered features and learned representations. It is central to the assignment because the two approaches do not only differ in performance; they also behave differently under distribution shift and offer different forms of explanation.

Slide 4 — Validation methodology

The validation strategy was designed to avoid leakage. The 60,000 training images were used for stratified five-fold cross-validation, while the 10,000 test images were kept untouched until both model configurations were frozen. The random seed was fixed at 123 for reproducibility. I used accuracy, macro precision, macro recall and macro F1 for evaluating the models. Macro F1 was especially important because it gives every class equal importance and can reveal weak classes such as Shirt, even when overall accuracy appears strong. This matters because a single train-validation split could give a lucky or unlucky estimate, while five folds show both average performance and stability across different partitions.

Slide 5 — Classical model

For the classical model, I used HOG followed by a Linear SVM. The HOG configuration used nine orientation bins, four-by-four pixel cells, two-by-two cell blocks and L2-Hys normalisation. This design is appropriate because Fashion-MNIST objects are mostly defined by silhouettes and edge structure. For the SVM, I tested C values of 0.1, 1 and 10. The best validation result came from C equals 0.1, meaning stronger regularisation generalised better. The selected HOG-SVM achieved a five-fold Macro F1 of about 0.889 and a final test Macro F1 of 0.8886. Its main advantage is simplicity: after feature extraction, the classifier is lightweight and interpretable as decision boundaries in HOG feature space.

Slide 6 — CNN architecture and training

The CNN was deliberately compact. It used two convolutional layers, two max-pooling operations, one dense hidden layer with dropout, and a final layer producing ten logits. I used

CrossEntropyLoss, Adam (Kingma and Ba, 2015) with a learning rate of 0.001, batch size 64 and dropout of 0.3. Early stopping monitored validation loss with patience of three epochs. The training curves show that the model learned quickly, while the validation loss became less stable after the best epoch. This is evidence that early stopping was useful for controlling overfitting. I avoided a large architecture because the assignment is about critical comparison, not maximising a benchmark through unnecessary complexity.

Slide 7 — Five-fold validation results

Across five stratified folds, the CNN achieved a mean Macro F1 of about 0.921, compared with about 0.889 for the HOG-SVM. This is an improvement of roughly 3.2 percentage points. The CNN also showed low fold-to-fold variation, with a standard deviation of around 0.0015 for Macro F1. The best epochs were 7, 7, 6, 6 and 8, which suggests reasonably stable convergence. These results justified freezing the CNN architecture rather than expanding the project into a large hyperparameter search.

Slide 8 — Final test comparison

After model selection was complete, both final models were trained on all 60,000 training images and evaluated once on the untouched test set. The HOG-SVM reached test accuracy of 0.8894 and Macro F1 of 0.8886. The CNN reached test accuracy of 0.9209 and Macro F1 of 0.9204. This confirms the cross-validation estimate. The CNN improved every class, but the largest gains were for visually similar upper-body garments such as Shirt, Coat and Pullover. This suggests that learned convolutional features captured discriminative details beyond the fixed HOG descriptor.

Slide 9 — Error analysis

The hardest class for both models was Shirt. The SVM achieved a Shirt F1 of 0.6895, while the CNN improved it to 0.7774. This is a meaningful gain, but Shirt remained much weaker

than categories such as Trouser, Bag or Sandal. The examples show why: at 28 by 28 resolution, a Shirt can look very similar to a T-shirt or Pullover. Therefore, some of the error comes from the model, but some also comes from genuine visual ambiguity in the dataset.

Slide 10 — Explainability

For explainability, I used LIME (Ribeiro, Singh and Guestrin, 2016) for the HOG-SVM and Integrated Gradients (Sundararajan, Taly and Yan, 2017) for the CNN. LIME is model-agnostic: it perturbs image regions and observes how the prediction changes. Integrated Gradients attributes the CNN's prediction back to input pixels using a baseline. I used these explanations diagnostically, especially for Shirt examples. In a misclassified Shirt, attribution focused on regions consistent with a T-shirt-like silhouette. I also attempted Grad-CAM (Selvaraju et al., 2017), but selected examples produced non-informative zero maps, so I documented the limitation rather than presenting misleading evidence. This is important because XAI outputs should be treated as evidence to interpret carefully, not as decoration.

Slide 11 — Robustness and diagnostic bias analysis

Fashion-MNIST does not contain demographic attributes, so demographic fairness cannot be evaluated. Instead, I investigated class disparity and image-quality disparity. Under dark and low-contrast transformations, the HOG-SVM performance remained unchanged because gradient directions are preserved and L2-Hys normalisation suppresses magnitude changes. However, under Gaussian noise, its Macro F1 collapsed to about 0.305 because noise creates artificial local gradients. The CNN was more sensitive to brightness and contrast changes, but retained much higher performance under noise, with F1 around 0.787. This shows that robustness is not one-dimensional: the classical model is robust to one kind of shift, while the CNN is robust to another.

Slide 12 — Ethical implications

Ethically, the risks depend on the deployment context. Fashion-MNIST has low direct privacy risk because it contains product-style images rather than identifiable people. But a real marketplace classifier may process seller or customer photos containing faces, homes or metadata, so privacy controls would still be necessary. Fairness must also be framed carefully. Since there are no demographic labels, I cannot claim demographic fairness. But unequal class performance can still have economic consequences. A seller whose products are often Shirts may experience more misclassification, reduced search visibility and lower user trust. The benchmark also excludes colour, cultural clothing, complex backgrounds and many real photographic conditions, so representation bias remains a major deployment concern.

Slide 13 — Deployment recommendation

Based on the evidence, I would deploy the CNN, but with safeguards. It achieved higher test performance, better Shirt classification and much stronger noise robustness. A realistic architecture would use an image-upload workflow, preprocessing, a CNN inference API and a confidence threshold. High-confidence predictions could be automatically assigned, while low-confidence cases should go to human review. The system should monitor class distribution, image statistics, low-confidence rates and human corrections. Retraining should be triggered by drift or verified performance decline, not done blindly. This deployment design links technical uncertainty to operational controls rather than assuming the model is always right.

Slide 14 — Conclusion

In conclusion, the CNN is the stronger model for this use case, with test Macro F1 of 0.9204 compared with 0.8886 for HOG-SVM. However, the evidence is bounded. Fashion-MNIST is grayscale, low resolution and limited to ten categories. It does not prove that the model will work equally well on diverse real-world product photos, cultural garments or customer uploads. The final recommendation is therefore nuanced: deploy the CNN with monitoring, confidence-

based review and retraining on representative real-world data, while recognising that HOG-SVM remains a useful lightweight baseline. The main learning outcome is that technical performance, interpretability, robustness and ethics must be considered together before selecting a model for deployment.

References

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